

Generation Interconnection System Impact Study Report

For Queue Project Q24-2394

Chaffee Gas Recovery 138 kV

May 04, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2394. The Interconnection Customer proposes a 833.0 MW wind facility interconnecting at the Chaffee Gas Recovery 138 kV substation in the Virginia area. The request was received on March 22, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 494 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,593,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of CHAFFEE GAS RECOVERY SUBSTATION to BLAINE 138 KV, the CHAFFEE GAS RECOVERY to MICHIGAN WIND 2 138 KV L loads to 282.8% of its 267 MVA rating. Project DFAX: 70.8%.

For the loss of MICHIGAN WIND 2 to GAINESVILLE 138 KV, the BLAINE to STONEWALL ENERGY CTR 138 KV loads to 236.8% of its 329 MVA rating. Project DFAX: 70.2%.

The CHAFFEE GAS RECOVERY to BLAINE 138 KV is overloaded to 223.0% of its 371 MVA emergency rating for the loss of the GAINESVILLE SUBSTATION to MICHIGAN WIND 2 138 KV. The project contributes 70.2% DFAX (584.5 MW) to this overload.

For the loss of CHAFFEE GAS RECOVERY to BLAINE 138 KV, the GAINESVILLE to NOVEC PARADISE 138 KV loads to 185.7% of its 390 MVA rating. Project DFAX: 70.8%.

The MICHIGAN WIND 2 to GAINESVILLE SUBSTATION 138 KV line is overloaded to 180.8% of its 419 MVA emergency rating for the loss of the CHAFFEE GAS RECOVERY to BLAINE 138 KV Line. The project contributes 70.8% DFAX (589.6 MW) to this overload.

STONEWALL ENERGY CENTER to FOUNTAIN 138 KV Line: 163.3% loading (496 MVA) following the outage of the GAINESVILLE to MICHIGAN WIND 2 138 KV; DFAX = 70.1%.

WEST STAUNTON to DEAN DRIVE AREA ELECTRIC GENERATORS 138 KV: 127.7% loading (440 MVA) following the outage of the CHAFFEE GAS RECOVERY to BLAINE 138 KV; DFAX = 37.2%.

The DEAN DRIVE AREA ELECTRIC GENERATORS to NOVEC PARADISE 138 KV is overloaded to 127.7% of its 440 MVA emergency rating for the loss of the CHAFFEE GAS RECOVERY to BLAINE 138 KV. The project contributes 37.2% DFAX (309.7 MW) to this overload.

The SCHOOLCRAFT to LAKERIDGE 138 KV line is overloaded to 122.4% of its 216 MVA emergency rating for the loss of the CUSTER to LOVING S 138 KV. The project contributes 5.8% DFAX (48.4 MW) to this overload.

WEST STAUNTON to WOODS DP 138 KV: 122.0% loading (574 MVA) following the outage of the BLAINE SUBSTATION to CHAFFEE GAS RECOVERY 138 KV; DFAX = 28.8%.

The ANGLETON to SCHOOLCRAFT SUBSTATION 138 KV is overloaded to 106.2% of its 369 MVA emergency rating for the loss of the CUSTER to LOVING S 138 KV. The project contributes 9.3% DFAX (77.9 MW) to this overload.

The ANGLETON to IDAHO SPRINGS 138 KV is overloaded to 105.6% of its 414 MVA emergency rating for the loss of the CUSTER to LOVING S 138 KV. The project contributes 9.3% DFAX (77.9 MW) to this overload.

The ELK HON to CLOVER HILL 138 KV L is overloaded to 101.7% of its 324 MVA emergency rating for the loss of the W STAUNTON to WOODS DP 138 KV. The project contributes 5.3% DFAX (44.5 MW) to this overload.

The CLOVER HILL to PRINCE WILLIAM DP 138 KV is overloaded to 101.7% of its 324 MVA emergency rating for the loss of the WOODS DP to WEST STAUNTON 138 KV L. The project contributes 5.3% DFAX (44.5 MW) to this overload.

The PEAVY FALL HY to SOUTH HOUSTON 138 KV is overloaded to 101.5% of its 507 MVA emergency rating for the loss of the CUSTER to LOVING S 138 KV. The project contributes 6.7% DFAX (56.0 MW) to this overload.

LUNA to PEAVY FALL HY 138 KV: 101.4% loading (573 MVA) following the outage of the CUSTER SUBSTATION to LOVING S 138 KV; DFAX = 6.7%.

The YAVAPAI to LUNA 138 KV Line is overloaded to 101.3% of its 623 MVA emergency rating for the loss of the CUSTER to LOVING SOUTH 138 KV. The project contributes 6.7% DFAX (56.0 MW) to this overload.

The ELK HORN to HOOVER 138 KV is overloaded to 100.8% of its 304 MVA emergency rating for the loss of the WOODS DP to W STAUNTON 138 KV. The project contributes 5.3% DFAX (44.5 MW) to this overload.

The HOOVER to MENAN 138 KV is overloaded to 100.8% of its 304 MVA emergency rating for the loss of the WEST STAUNTON to WOODS DP 138 KV. The project contributes 5.3% DFAX (44.5 MW) to this overload.

The MENAN SUBSTATION to NORFORK 138 KV Line is overloaded to 100.8% of its 304 MVA emergency rating for the loss of the WOODS DP to W STAUNTON 138 KV. The project contributes 5.3% DFAX (44.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHAFFEEGAS RECOVERY to MICHIGAN WIND 2 138 KV	Rebuild line with higher-capacity conductor	\$1,657,000	98% (\$1,625,000)
BLAINE to STONEWALL ENERGY CTR 138 KV	Rebuild line with higher-capacity conductor	\$2,193,000	99% (\$2,162,000)
BLAINE to CHAFFEE GAS RECOVERY 138 KV	Rebuild line with higher-capacity conductor	\$2,717,000	99% (\$2,678,000)
GAINESVILLE to NOVEC PARADISE 138 KV	Rebuild line with higher-capacity conductor	\$2,121,000	98% (\$2,080,000)
MICHIGAN WIND 2 to GAINESVILLE SUBSTATION 138 KV line	Rebuild line with higher-capacity conductor	\$1,363,000	98% (\$1,337,000)
STONEWALL ENERGY CENTER to FOUNTAIN 138 KV Line	Rebuild line with higher-capacity conductor	\$1,706,000	94% (\$1,603,000)
WEST STAUNTON to DEAN DRIVE AREA ELECTRIC GENERATORS 138 KV	Rebuild line with higher-capacity conductor	\$840,000	87% (\$729,000)
DEAN DRIVE AREA ELECTRIC GENERATORS to NOVEC PARADISE 138 KV	Rebuild line with higher-capacity conductor	\$2,572,000	87% (\$2,233,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SCHOOLCRAFT to LAKERIDGE 138 KV line	Rebuild line with higher-capacity conductor	\$2,528,000	16% (\$391,000)
WEST STAUNTON to WOODS DP 138 KV	Rebuild line with higher-capacity conductor	\$798,000	59% (\$472,000)
SCHOOLCRAFT to ANGLETON 138 KV	Reconductor line	\$2,011,000	15% (\$305,000)
ANGLETON to IDAHO SPRINGS 138 KV	Reconductor line	\$1,461,000	15% (\$221,000)
ELK HORN SUBSTATION to CLOVER HILL 138 KV	Reconductor line	\$337,000	42% (\$141,000)
CLOVER HILL to PRINCE WILLIAM DP 138 KV	Reconductor line	\$944,000	42% (\$395,000)
S HOUSTON to PEAVY FALL HY 138 KV	Reconductor line	\$515,000	12% (\$59,000)
LUNA to PEAVY FALL HY 138 KV	Reconductor line	\$732,000	12% (\$84,000)
YAVAPAI to LUNA 138 KV Line	Reconductor line	\$581,000	12% (\$67,000)
ELK HORN to HOOVER 138 KV Line	Reconductor line	\$416,000	42% (\$174,000)
HOOVER to MENAN 138 KV	Reconductor line	\$465,000	42% (\$195,000)
NORFORK to MENAN 138 KV Line	Reconductor line	\$1,534,000	42% (\$642,000)
	Total allocated to project		\$17,593,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.